Batterjee Medical College
For Science and Technology

BMC

Tubercelosis

An approach to a Case of Hemoptysis

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BATTERJEE MEDICAL COLLEGE

Learning Objectives

Knowledge:

- Describe the epidemiology, reservoirs, and modes of transmission of Mycobacterium tuberculosis and other species in the Mycobacterium tuberculosis complex.
- Understand the mechanisms of primary TB, latent TB infection (LTBI), active reactivation, and tissue destruction.
- Learn the clinical presentation and physical findings of both pulmonary TB and extrapulmonary TB variants.
- Understand the diagnostic modalities and evidence-based treatment regimens for latent, active, drug-susceptible, and **drug-resistant TB.**

Skills:

- Interpret clinical presentations, physical examination findings, laboratory values and imaging to reach a reasonable diagnosis.
- Apply evidence-based management plans for initiating initial and continuation phase anti-TB therapy, managing treatment interruptions, and utilizing adjunctive therapies.

Values

- Demonstrate a professional attitude in strict adherence to the training site's rules and regulations.
- Prioritize public health safety, mandatory case reporting, airborne precautions, and contact tracing to reduce transmission risks.

Main Subtopics

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- Etiology
- Classification
- Pathogenesis
- Clinical Manifestations
- Diagnostics
- Treatment
- Subtypes and Variants
- Complications

Tuberculosis

Definition:

A disease caused by *Mycobacterium tuberculosis*, which typically affects the lungs.

Primary infection, transmitted via airborne aerosol (droplet), is often initially asymptomatic.

Latent infection, *M. tuberculosis* infection is typically dormant, because of intact innate and cellular immune responses. If the immune system is compromised, reactivation of the infection may occur and lead to active TB.



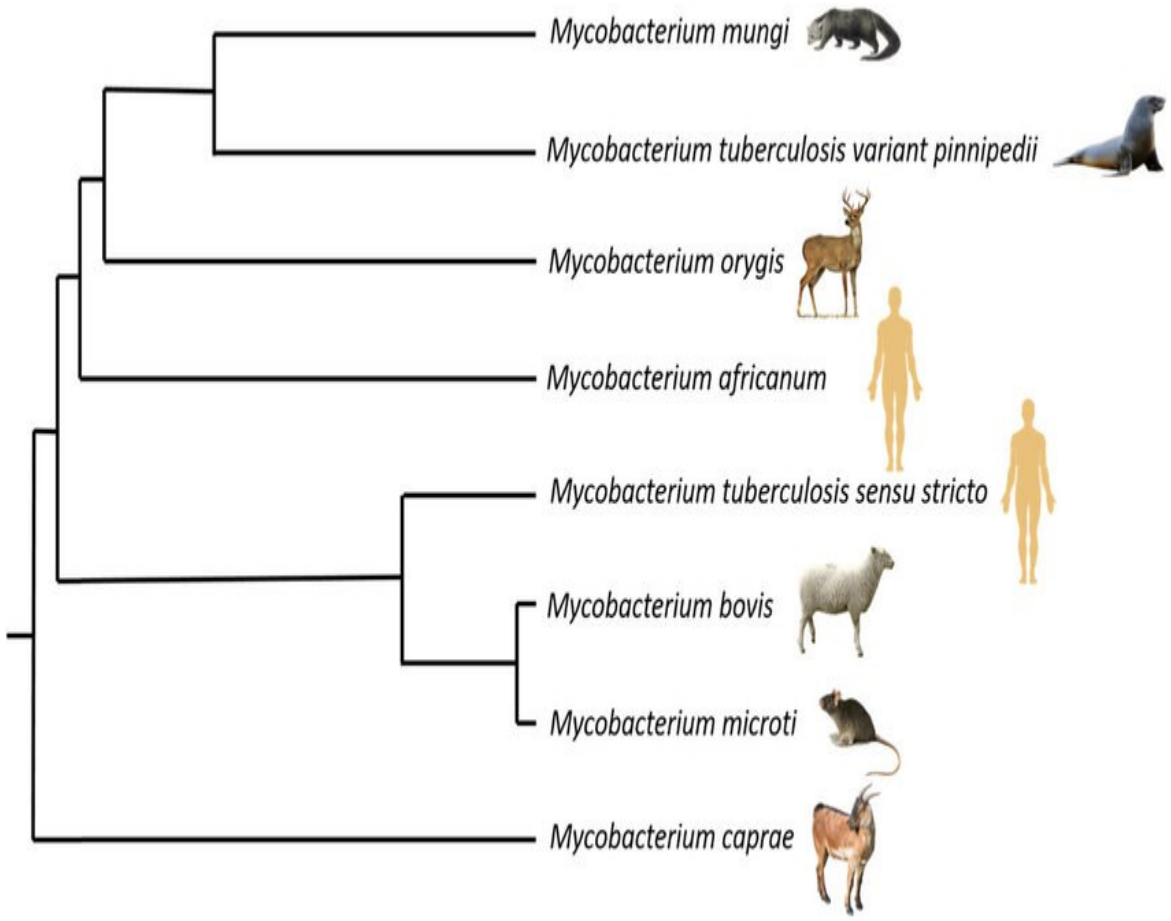
Tuberculosis

Etiology

Mycobacteria Species

Mycobacterium species that cause tuberculosis are collectively known as the *Mycobacterium tuberculosis* complex, which includes:

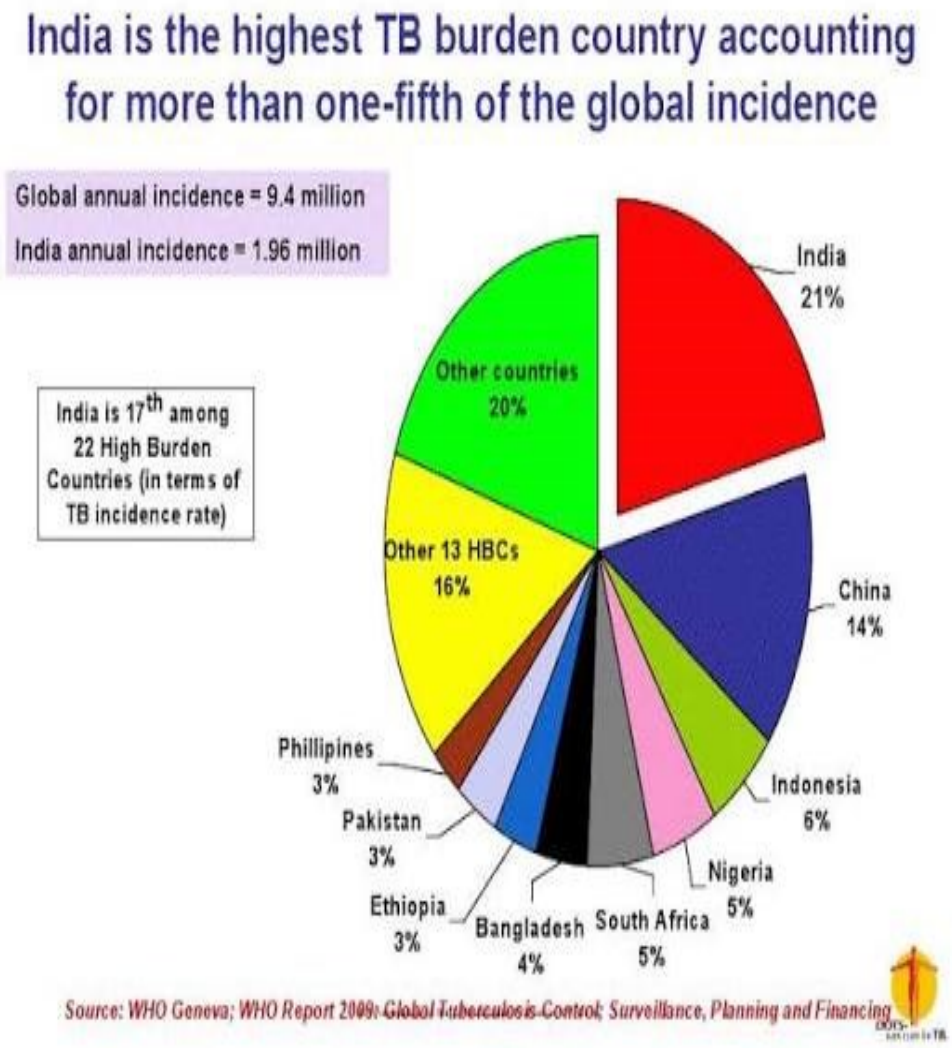
- ***Mycobacterium tuberculosis***
  - Mode of transmission: spread via aerosol droplet nuclei
  - Reservoir: predominantly humans
- ***Mycobacterium bovis***
  - Mode of transmission: ingestion of contaminated cow's milk
  - Reservoir: predominantly cattle
- ***Mycobacterium africanum***: common cause of tuberculosis in West, Central, and East Africa
- ***Mycobacterium microti***



Tuberculosis

Risk factors for tuberculosis exposure

- Working in the health care industry
- Migration from countries with a high TB incidence
- Frequent travel to countries with a high TB burden
- Close contact with a patient with active TB infection
- Crowded living conditions (e.g., prisons)
- Homelessness
- Smoking
- Associated disease (HIV, Renal failure, DM)



Tuberculosis

Pathophysiology

Primary tuberculosis

Innate immune response

•Exposure to *M. tuberculosis*

- Inhaled droplet nuclei reach the terminal alveoli and are taken up by the alveolar macrophages.
- Alveolar macrophages recognize *M. tuberculosis* and Phagocytosis of mycobacteria.
- *M. tuberculosis* survives within macrophages

•Macrophage lysis and release of bacteria

- Following replication in the alveolar macrophages, the released bacteria attack uninfected macrophages to spread infection.
- Some bacteria enter the bloodstream, causing bacteremia and affects multiple organs.

Tuberculosis

Pathophysiology

•Granulomatous inflammation and tissue destruction

- Destruction of *M. tuberculosis* infected macrophages causes central caseous necrosis and tissue damage.
- Granuloma limits the spread of infection.
- **Ghon focus**: a granuloma typically located in the middle/lower lung lobes.
- **Ghon complex**: formed by the Ghon focus, regional lymph node, and the linking lymphatic vessels

•Disease progression

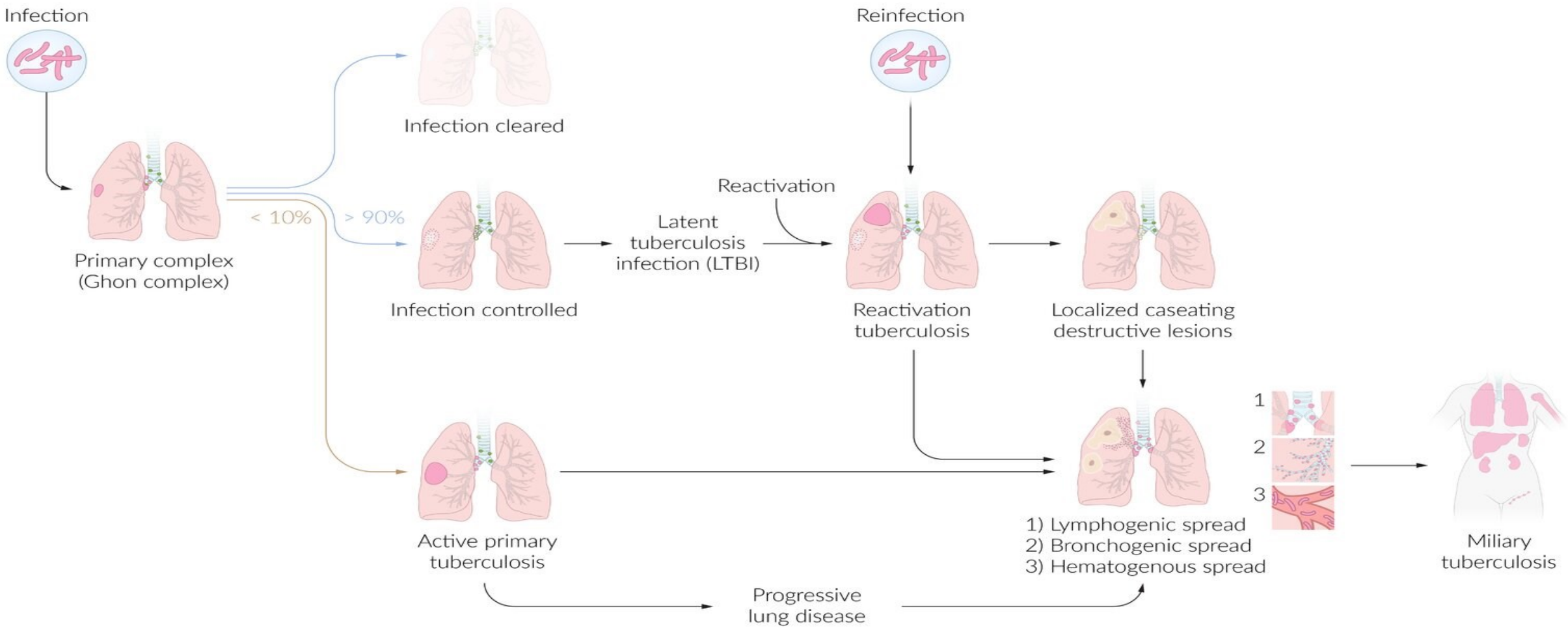
- Sufficient immune response
  - Most of the bacteria are killed.
  - Some bacteria may persist, causing Latent tuberculosis.
  - The granulomas in the Ghon complex undergo fibrosis and calcification
- Deficient immune response (e.g., HIV, malnutrition)
  - Failure of granulomas to limit infection
  - **Progressive primary TB** causing progressive lung disease, bacteremia, and miliary TB

Tuberculosis

Secondary tuberculosis

- **Latent TB**: Dynamic equilibrium is maintained between the host immune response and *M. tuberculosis*.
- **Reactivation of disease**: due to weakening of immune response
- **Disease progression**
  - Caseating granulomas with central necrosis and Langhans giant cells are characteristic features.
  - Usually affects the upper lobes of the lungs because of higher oxygen tension
  - Can also affect other organs
  - Previous sensitization to mycobacterial antigens results in a stronger inflammatory response, causing extensive tissue destruction, cavitation, and scarring.

Tuberculosis



Tuberculosis

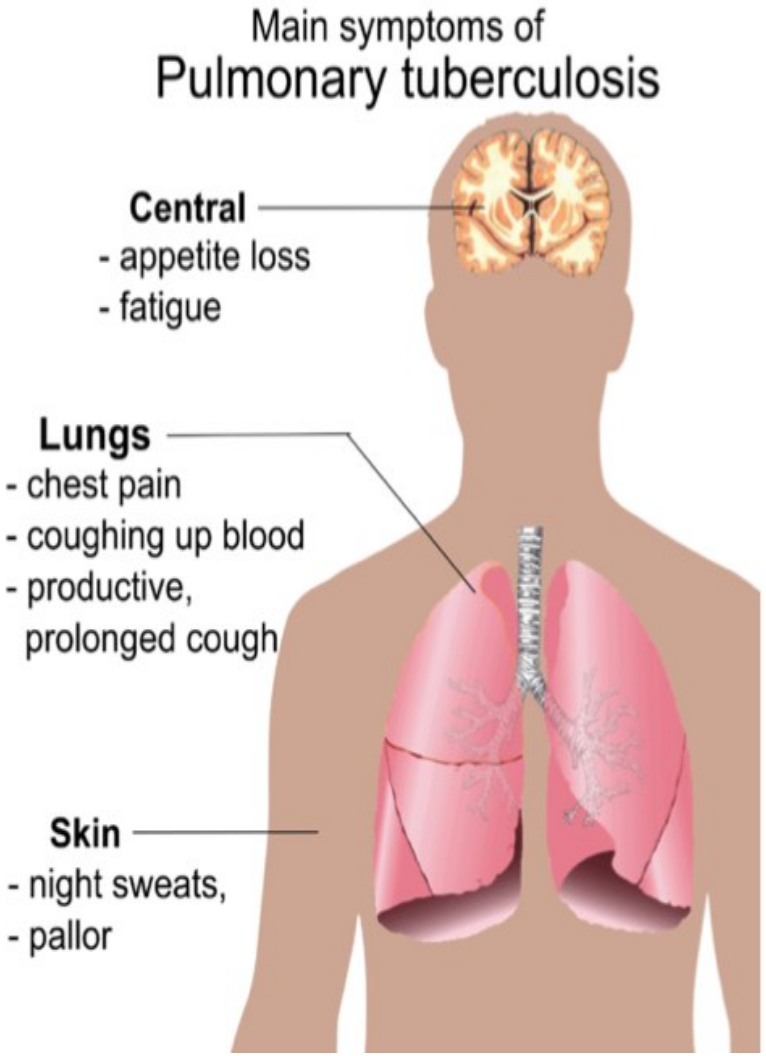
Pulmonary tuberculosis

•Systemic

- Low-grade fever with night sweats
- Weight loss (often severe), anorexia
- Decreased appetite
- Malaise, weakness

•Pulmonary

- Non-productive cough
- Productive cough with purulent sputum, hemoptysis
- Shortness of breath
- Pleuritic chest pain

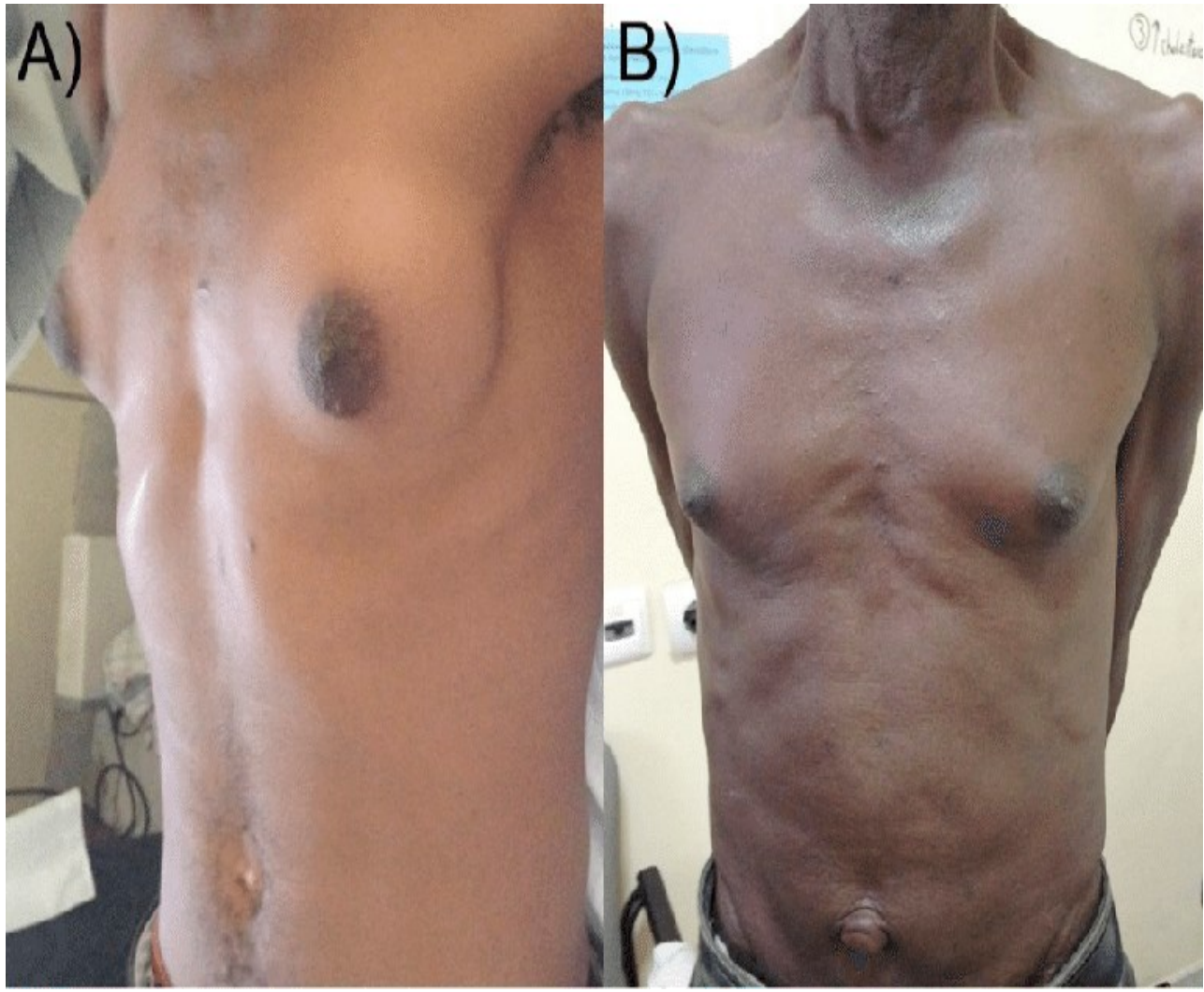


Tuberculosis

Pulmonary tuberculosis

•**Clinical examination**: Findings are nonspecific.

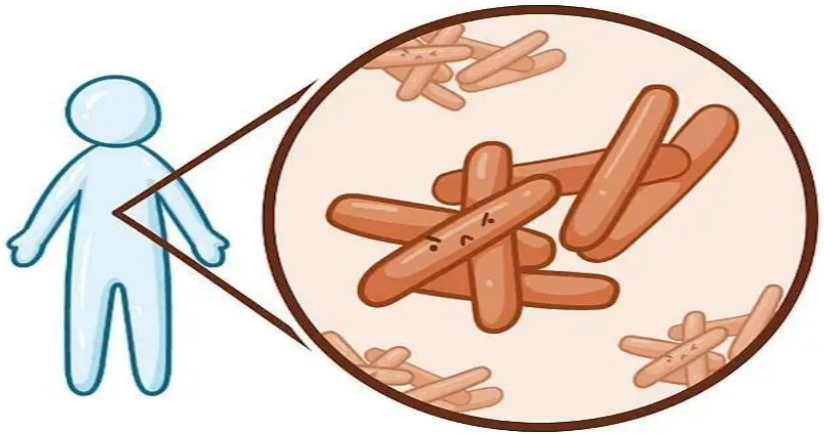
- General: pallor, clubbing, generalized wasting
- Chest examination: Findings vary depending on the type and degree of pulmonary involvement.
  - Consolidation: dullness on percussion, crackles, diminished breath sounds
  - Cavitation: hyperresonance on percussion, bronchial breathing
  - Rhonchi
  - Pleural effusion and/or empyema: dullness on percussion, diminished breath sounds
- Other possible findings: erythema nodosum, conjunctivitis



Tuberculosis

LATENT TB

ASYMPTOMATIC CONDITION



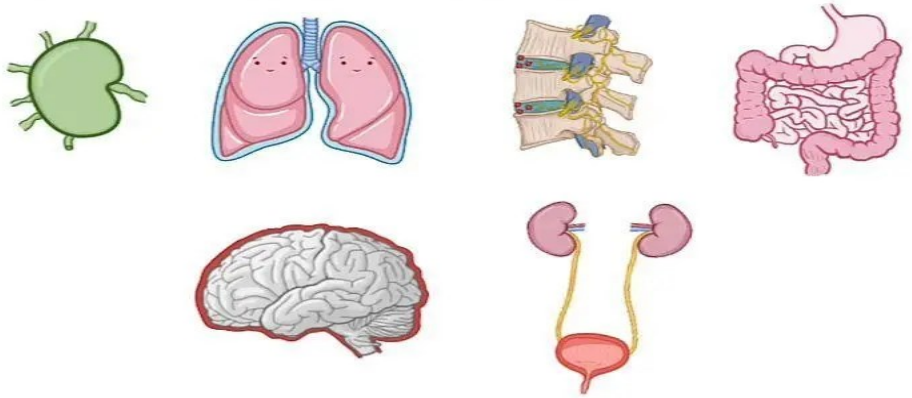
MYCOBACTERIUM TUBERCULOSIS

HELD in CHECK by IMMUNE SYSTEM

EXTRAPULMONARY TB

AFFECT ORGANS other than LUNGS

* TUBERCULOUS LYMPHADENITIS
* PLEURAL TB
* SPINAL TB
* TUBERCULOSIS PERITONITIS
* TUBERCULOSIS MENINGITIS
* GENITOURINARY TB

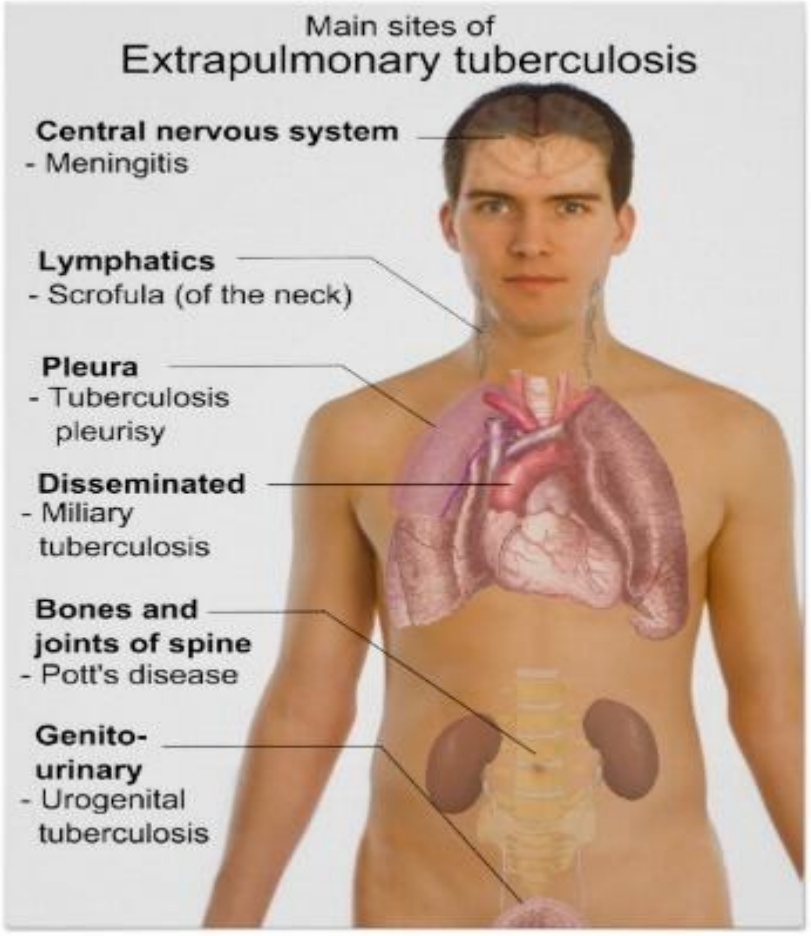


Tuberculosis

Extrapulmonary tuberculosis

Tuberculous lymphadenitis

- **Sites of lymph node involvement**
  - Cervical, Inguinal, Axillary
- **Clinical features**
  - Constitutional symptoms
  - Lymphadenopathy
    - Lymphadenitis firm, mobile, and discrete lymph node
    - Periadenitis: rubbery and fixed lymph nodes
    - Cold abscess: soft and fluctuant lymph nodes



Tuberculosis

Extrapulmonary tuberculosis

Tuberculous pleurisy

•Pathophysiology

- Direct spread of infection from the lungs
- Hematogenous spread
- Delayed hypersensitivity reaction in the pleural space due to mycobacterial antigen

•Clinical features

- Constitutional symptoms
- Nonproductive cough
- Intensive (pleuritic) chest pain
- Dyspnea

Tuberculosis

Extrapulmonary tuberculosis

Miliary tuberculosis

•**Definition**: a form of TB with multiorgan involvement that manifests with granulomatous lesions, as a result of lymph or blood spread of bacilli from a pulmonary or extrapulmonary source

•**Common sites of involvement**: lungs, spleen, liver, lymph nodes, adrenals, meninges, vertebrae (Pott disease), joints and long bones, and choroid

•**Clinical features**: mostly nonspecific

- Cough and dyspnea
- Constitutional symptoms
- Lymphadenopathy
- Hepatosplenomegaly
- Fundus may show choroidal tubercles

Tuberculosis

Extrapulmonary tuberculosis

Tuberculous meningitis

•Pericardial tuberculosis

•Types

- Pericardial effusion (rarely, tamponade)
- Constrictive pericarditis

•Clinical features

- Constitutional symptoms
- Chest pain
- Cough
- Dyspnea

Tuberculosis

Extrapulmonary tuberculosis

Cutaneous tuberculosis

Diagnostics: skin biopsy for histopathology, acid-fast staining, and culture

Gastrointestinal tuberculosis

•Pathophysiology

- Ingestion of infected milk or sputum
- Hematogenous or lymphatic spread
- It can affect Peritoneum, esophagus, stomach, jejunum, ileocecum, & colorectal region

Tuberculosis

Extrapulmonary tuberculosis

Urogenital tuberculosis

•Pathophysiology

- Hematogenous spread of infection to the kidneys
- Ureters and bladder are infected when bacteria are excreted in urine.

•Clinical features

- Constitutional symptoms
- Flank pain
- Hematuria
- Storage symptoms
  - Increased frequency of voiding
  - Nocturia
  - Urgency
- Hypertension
- Recurrent UTIs unresponsive to standard antibiotic therapy

Tuberculosis

Extrapulmonary tuberculosis

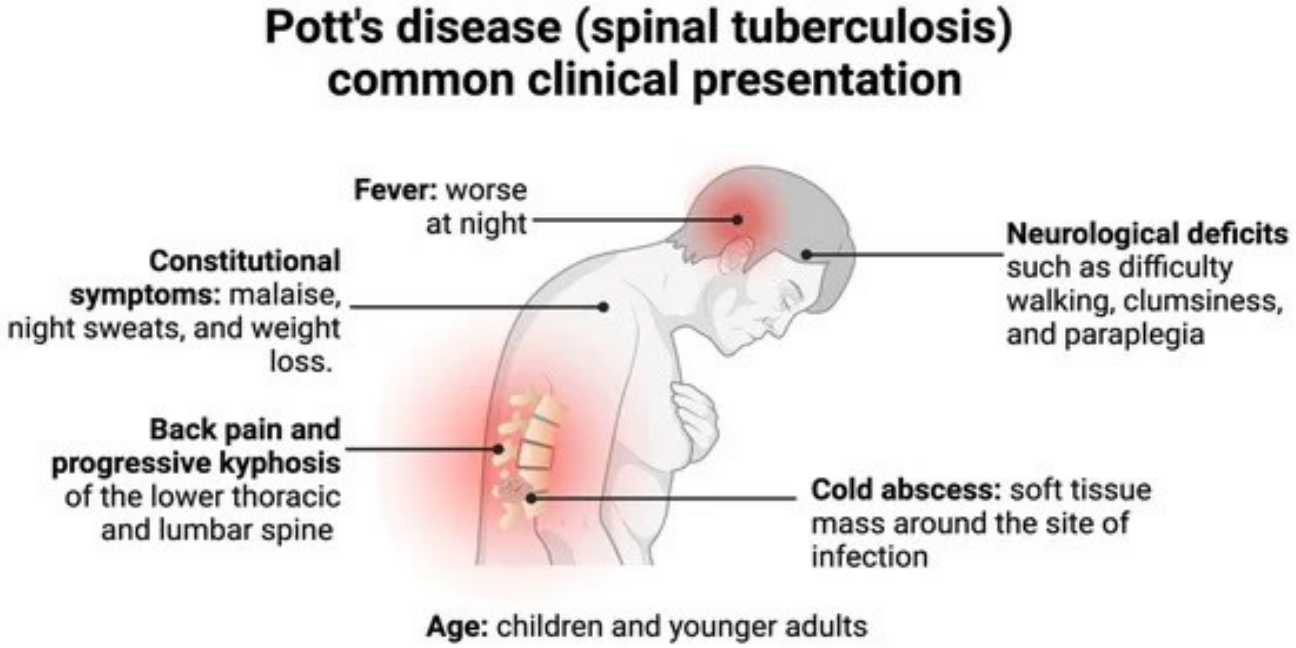
Pott disease

•Pathophysiology

- Hematogenous dissemination of infection to vertebral bodies following primary TB
- Infection from vertebral bodies spreads to the intervertebral disk, causing destruction and collapse of the disk

•Clinical features

- Constitutional symptoms
- Back pain
- Spinal tenderness
- Kyphosis
- Deformity
- Neurological deficit (e.g., paraplegia)

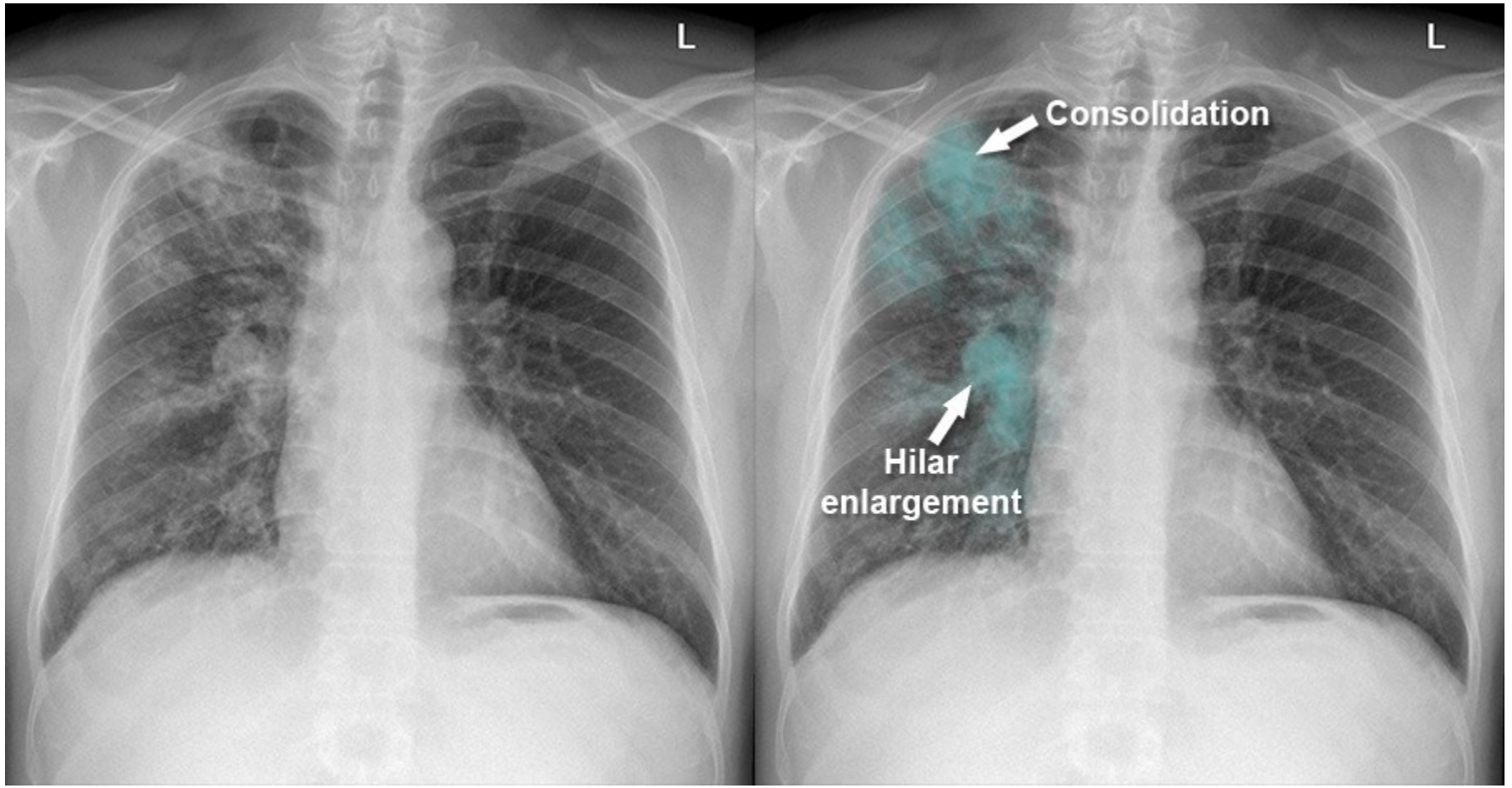


Tuberculosis

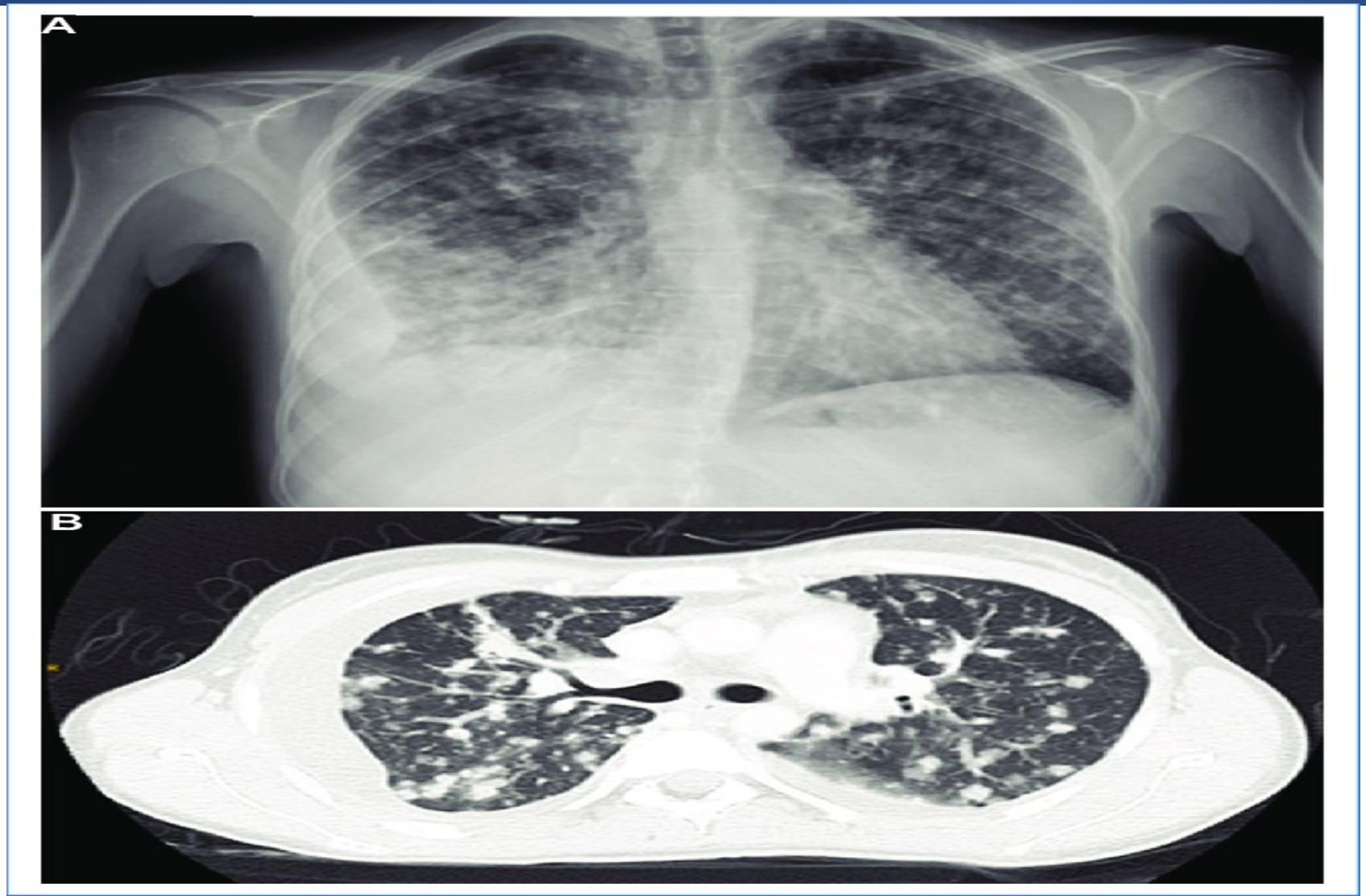
Diagnosis

- **Initial workup in all patients**
  - History (e.g., exposure to individuals with TB, history of visiting or living in countries with a high TB burden)
  - Microbiological studies
    - Initial tests: microscopy examination for acid-fast bacilli, nucleic acid amplification test
    - Confirmatory test: mycobacterial culture
    - Drug susceptibility testing: indicated in certain circumstances to identify resistant mycobacteria
  - Consider CT chest or CXR
- **Suspicion for extrapulmonary TB**: Additional invasive procedures (e.g., lymph node biopsy, thoracocentesis) may be required for bacteriological confirmation

Tuberculosis



Tuberculosis



Tuberculosis

Laboratory studies

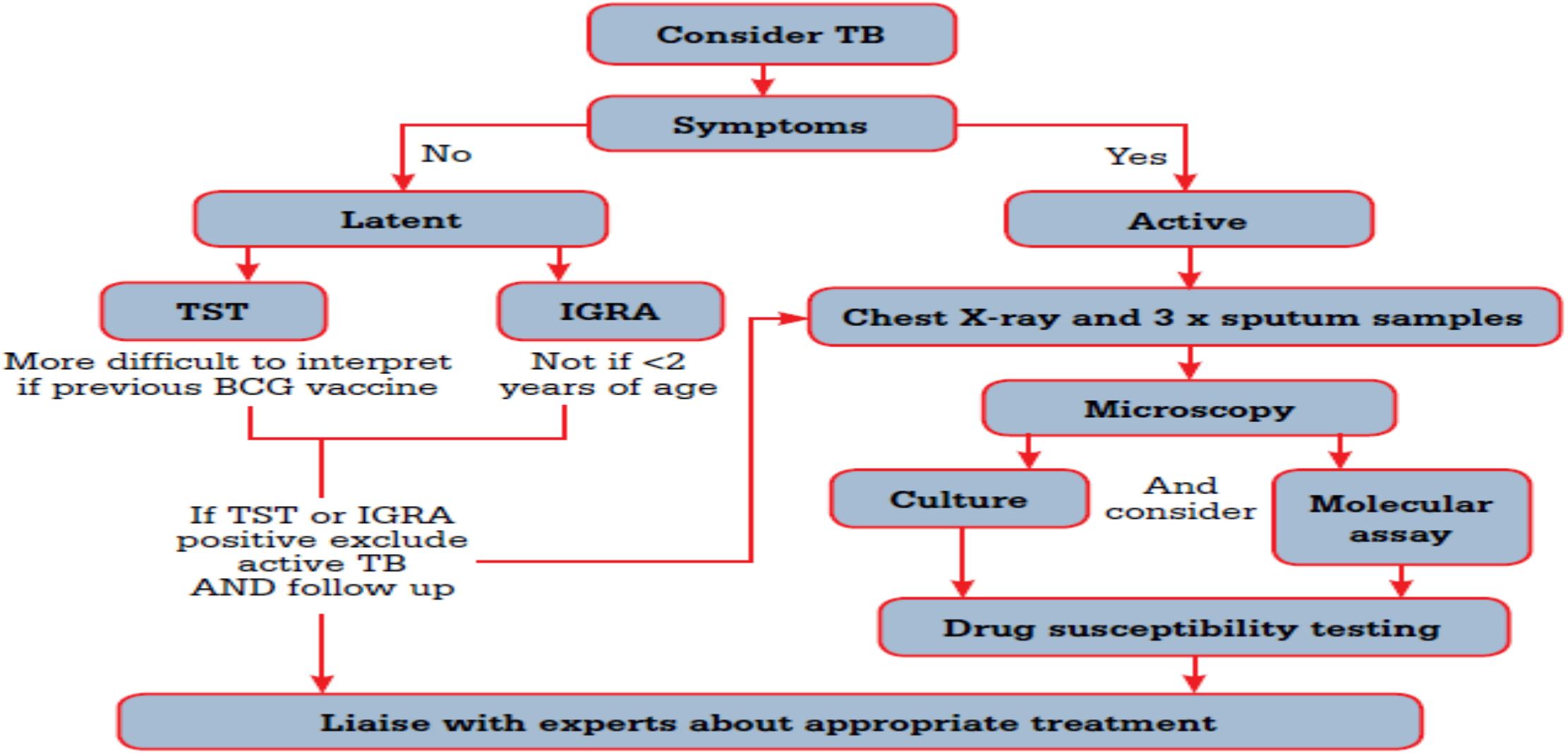
Unspecific, provide important information during initial diagnostic stages.

- **CBC**: mild leukocytosis (common) , anemia
- **Sodium**: mild hyponatremia
- **Liver enzymes**: ↑ in concomitant liver disease; useful as a baseline prior to treatment
- **Other nonspecific findings**: ↑ ESR, pyuria in renal disease
- Interferon gamma release assay
- Serum QuantiFERON test
- **Microbiological studies**

Confirmation of the presence of bacilli in different samples is done by direct visualization, positive culture, or by detecting genetic material.

- Sputum samples: Obtain ≥ 3 samples separated by 8 to 24-hour intervals (should include at least one early morning sample).
- Induced sputum
- Gastric lavage
- Bronchoalveolar lavage
- Extrapulmonary TB suspected: Fluid specimens or tissue samples should be sent for cell count, chemistries, AFB smear microscopy, and adenosine deaminase

Tuberculosis



Tuberculosis

Treatment

General principles

•Goals

- Reduction of disease severity and risk of transmission
- Eradication of all bacilli to achieve sustained cure without relapse following completion of treatment
- Prevention of drug resistance during therapy

•Infection control measures

- Case notification: mandatory reporting to local health department
- Airborne precautions: should include a surgical mask for the patient and adequate personal protective equipment (including respirators) for medical staff and caregivers
- Contact tracing

Tuberculosis

Antituberculosis therapy*:*

RIPE TB regimen

•**Intensive phase**: 8 weeks (~ 2 months) of rifampin PLUS isoniazid, pyrazinamide, and ethambutol

•**Continuation phase**: 18 weeks (~ 4 months) of rifampin, PLUS isoniazid

•**Additional treatment**: pyridoxine to prevent vitamin B_6 deficiency resulting from isoniazid (promotes pyridoxine excretion) for all individuals at risk of neuropathy

•**Additional considerations**

- Self-administered and daily dosages are preferred
- Directly observed therapy (DOT; standard practice): Health care personnel observe the patient taking the medication.
- Restart full treatment if there is an interruption of ≥ 14 days during the intensive phase.

Tuberculosis

Antituberculosis therapy*:*

4-month regimen

Used for patients with drug-susceptible pulmonary TB who are age ≥ 12 years, ≥ 40 kg, and without contraindications (e.g., drug-drug interactions, prolonged QT syndrome, pregnancy).

- **Intensive phase**:

8 weeks of rifapentine, isoniazid, pyrazinamide, and moxifloxacin

- **Continuation phase**:

9 weeks of rifapentine, isoniazid, and moxifloxacin

- **Adjuvant treatment**: Pyridoxine to prevent vitamin B_6 deficiency resulting from isoniazid

Tuberculosis

Antituberculosis therapy*:*

Treatment of drug-resistant TB disease

•**Agents**

- Oral: fluoroquinolone (levofloxacin or moxifloxacin), linezolid, clofazimine, cycloserine,
- IV: amikacin, streptomycin, and carbapenem with amoxicillin-clavulanic acid

•**Phases of treatment**: Combine at least 5 drugs for the intensive phase, and at least 4 drugs for the continuation phase.

•**Role of surgery**: can be beneficial.

Tuberculosis

Antituberculosis therapy*:*

Treatment of extrapulmonary TB

•**Standard antituberculosis therapy**: used for lymph node, pleural, miliary, pericardial, adrenal, gastrointestinal, and genitourinary TB

•**Extended regimens**

- TB meningitis
  - 2 months of 4-drug intensive phase, followed by 7–10 months of continuation phase with isoniazid and rifampin
  - Adjunctive glucocorticoid treatment with dexamethasone or prednisolone for 6–8 weeks
- Pott disease and other bone and joint TB: treatment duration of at least 9 months; surgery may be necessary.

Tuberculosis

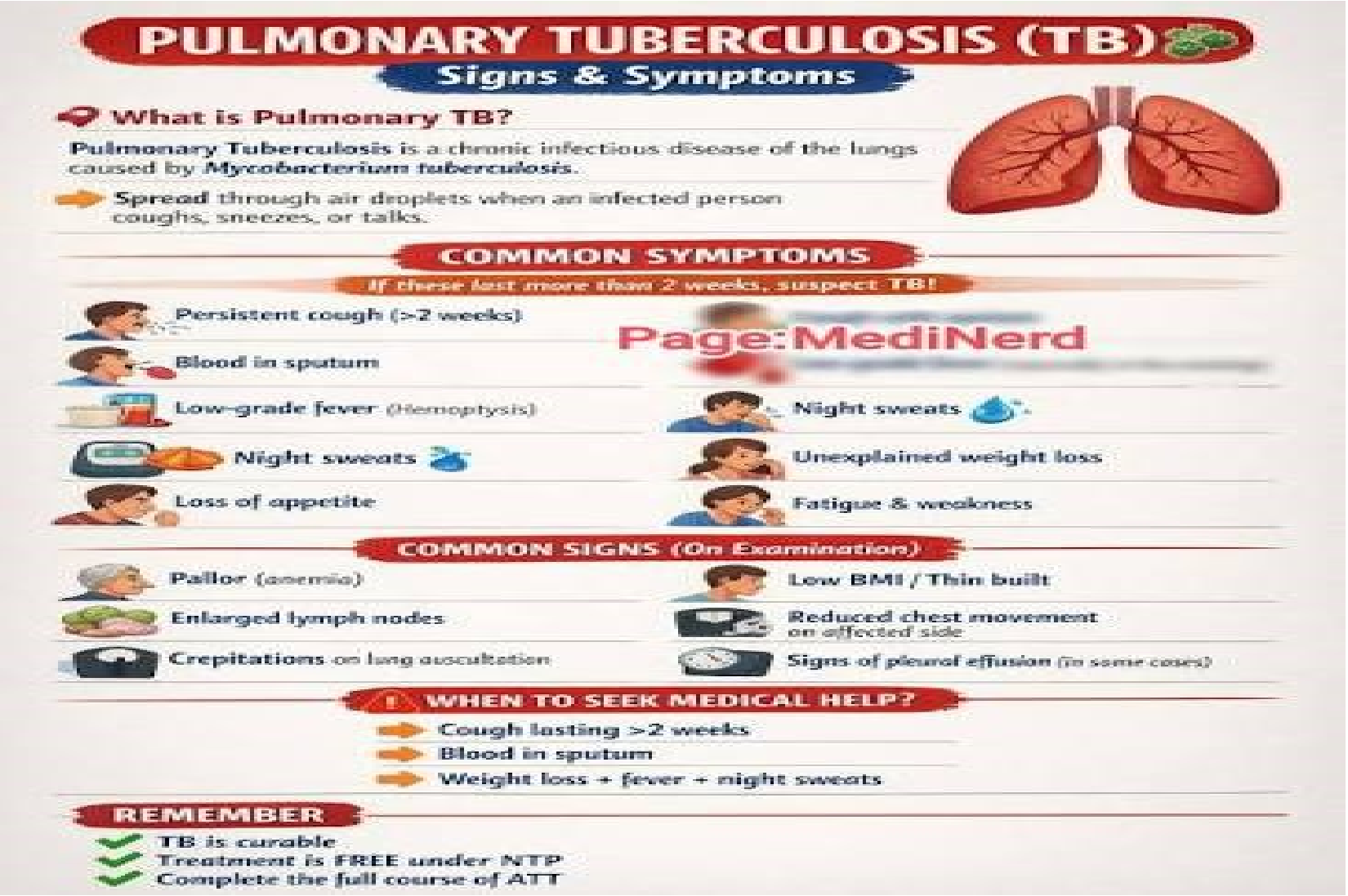
Treatment of latent TB:

•Pharmacological therapy: indicated for patients with positive IGRA or TST after active TB has been excluded

Rifamycin-based short regimens are preferred.

Short Term regimen
• Once-weekly isoniazid PLUS once-weekly rifapentine for 3 months • PLUS once-weekly pyridoxine
• Once-daily isoniazid PLUS once-daily rifampin for 3 months • PLUS once-weekly pyridoxine

Summary



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Case Scenario 1

A 48-year-old man comes to the physician for a preplacement examination and tuberculosis test before starting a volunteer job in a refugee clinic. He has had a mild cough, sore throat, and occasional headaches for 1 week. He has hypertension and gastroesophageal reflux disease. He works as a mechanic and has never traveled outside of the US. He lives alone and has had no known contact with anyone infected with tuberculosis. He has smoked one pack of cigarettes daily for 27 years and does not drink alcohol. Current medications include ramipril and esomeprazole. He is 180 cm (5 ft 11 in) tall and weighs 89 kg (196 lb); BMI is 28 kg/m^2. His temperature is 37.3°C (99.1°F), pulse is 81/min, respirations are 17/min, and blood pressure is 145/75 mm Hg. Pulmonary examination shows no abnormalities. Laboratory studies show a leukocyte count of 6200/mm^3 and a hemoglobin concentration of 13.8 g/dL. A rapid test for COVID-19 is negative. A tuberculin skin test (TST) shows an induration of 13 mm in diameter after 50 hours. Which of the following is the most appropriate next step in management?

A. Chest x-ray
B. No further management is indicated
C. Interferon-γ release assay
D. Repeat tuberculin skin test after 8–10 weeks
E. Isoniazid and rifapentine therapy

Case Scenario 2

A 7-year-old boy is brought to a refugee clinic by his parents. He has not seen a physician since leaving Ukraine with his family 2 months ago. His parents report that he has been healthy except for occasional episodes of nasal discharge and sneezing when playing outside. Medical history is remarkable for infantile atopic dermatitis treated with emollients. He received all appropriate childhood vaccinations in Ukraine, including the BCG vaccine, as Ukraine is considered a country with a high burden of tuberculosis. He has two healthy siblings. His mother is a physician and his father works in agriculture. Vital signs are within normal limits. The lungs are clear to auscultation. The remainder of the physical examination shows no abnormalities. Which of the following is the most appropriate next step in management?

A. Tuberculin skin test
B. No further evaluation is necessary
C. X-ray of the chest
D. Sputum microscopy and culture
E. Interferon-gamma release assay

Questions?



References & further reading

- Step-up to medicine 5th edition
- Harrison's Principles of Internal Medicine
  20th edition
- Amboss
- Cecil Essentials of Medicine 10th edition
- First Aid for the internal medicine board
  4th edition
- Lancet
- Davidson's principles & practice of medicine 21st edition

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Thank You